

Robust Optimized Pulse Schemes for Atomic Fountain Interferometry

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Abstract: The robustness of an atomic fountain interferometer with respect to variations in the signal velocity of the atoms and detuning from the optimal pulse sequence is examined. We numerically optimize the detuning of an atom interferometer sequence with a sequence separation of 2.5 ns and compare the expected signal contrast resulting on the one hand from the signal velocity fluctuations and the value of the laser field amplitudes. The atom detuning results in a phase (area) error that produces significant fluctuations in the expected signal contrast. The results show that the sequence separation of 2.5 ns is not optimal for the atom interferometer. The atom detuning results in a phase (area) error that produces significant fluctuations in the expected signal contrast. The results show that the sequence separation of 2.5 ns is not optimal for the atom interferometer. The atom detuning results in a phase (area) error that produces significant fluctuations in the expected signal contrast. The results show that the sequence separation of 2.5 ns is not optimal for the atom interferometer.

Keywords: atom interferometry; pulse sequence; optimal control; numerical optimization

1. Introduction

Atom interferometry [1–5] is a new technology for quantum metrology and quantum control [6]. The basic principle exploits the wave nature of matter in the same way that classical optical interferometers exploit the wave nature of light. In the type of atom interferometry considered here, an initial wavepacket is split into two paths that are separated in time. These paths are then recombined and measured. The combination of the separated paths differs from the two paths in the propagation of the paths, as well as the resulting pattern.

For atoms, the paths are affected by gravitational fields, accelerations, and rotations. This enables the only applications in fundamental physics [7–9], but also direct practical implementations of gravimetry, gyroscopy, and inertial navigation [10]. A second advantage, as well as a challenge of atom interferometry, is that the wavepacket is many orders of magnitude smaller than that of light, thus promising extremely high sensitivity, but making it difficult to implement “beam splitters” and “mirrors” that are different at that scale. The highest precision can be reached through light–matter interaction [11–13], specifically, the interaction of the atoms with counter-propagating laser fields. There are different regimes in which the atoms can absorb or emit photons, imparting momentum kicks [14]. Most commonly, in the Raman regime the change in momentum is associated with a change in the internal state of the atom [15–17], whereas in the Bragg regime the atom remains in the internal ground state [18–21].

Thus, we consider the setup of an atomic fountain interferometer (AFI) [22–24]. We begin in the Bragg regime, which allows for higher momentum space separation and is less susceptible to technical and atomic shifts [25]. A cloud of atoms with well-defined initial momentum is launched into a 3D optical trap [26]. After the launch, counter-propagating laser beams with appropriately controlled amplitude and phase impart momentum kicks onto the atoms to implement a momentum space manipulation, mirror, and recombiner.



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